

JioMausam Drill book

What we changed and why, the final deck slide by slide, every cross-question we can think of, and what we do if an assumption fails.

How to use it: read Parts A and B once to know the story. Drill Part C out loud: say the answer, then stop. Know Part D well enough to answer “what if this fails?” for any slide. Keep Part E open the night before.

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How the idea changed, and why

Our submitted video pitched an insurance-first idea: Jio's towers measure rain, and Mausam Kavach pays riders automatically. Stress-testing broke three things: **accuracy** (links alone caught 68% of rain events), **who pays** (unproven), and **evidence** (several claims we could not back). The final deck keeps the core (Jio's network signal and the Mumbai problem) but leads with weather intelligence and makes insurance optional.

THE LINE TO OWN THE CHANGE

"Our original idea was Kavach. After stress-testing the accuracy and who pays, we realised the stronger business is the weather intelligence underneath it. Kavach becomes one application of that intelligence."

What the video said → what we say now → why

THE VIDEO SAID	NOW WE SAY	WHY WE CHANGED IT
"India's biggest weather network already exists... lakhs of stations... every tower is a rain gauge."	"Jio's microwave links may carry a weather signal."	Only microwave links react to rain; fibre does not. Nobody outside Jio knows how many usable links it has, so "lakhs" and "every tower" were claims we could not back. Coverage is now Gate 1.
"Our rain readings matched national radar at 0.95."	"Links tracked radar closely, but alone caught only 68% of rain events. Our bar is 80%."	0.95 is how well the <i>network average</i> follows radar hour by hour; local errors cancel out, so it flatters. What matters for alerts and payouts is how many real events are caught.
"Our payout rule made the same decision as radar 95% of the time."	"68% of rain events caught." (94.6% kept for Q&A only)	94.6% agreement is mostly dry days where both said "no". It is true but misleading as a headline.
"We built it... on 500 real microwave links."	"500 real links from a public dataset in Germany, not Jio's ."	The old wording could make a judge think we had Jio data. We never did.
"A live, street-level weather map... minute by minute."	"Decision infrastructure, not a weather map"; "spatial weather grid"; "near-real-time".	Resolution depends on link density and refresh rate depends on telemetry; both are pilot questions. And nobody pays for a map; they pay for better decisions.
"Delivery apps buy Kavach for riders... ₹300 lands in the rider's JioPay wallet within hours."	"Insurance is one application, not the anchor. A licensed insurer would underwrite and pay." ₹300 = prototype assumption.	Who pays is unproven, Jio cannot carry insurance risk, links alone are not accurate enough to pay claims on, and the payment rail is the insurer's choice.
"50,000 riders ≈ ₹12 crore of premiums; 1% of Jio users = ₹174 crore."	Removed from the deck. Q&A only, always called a <i>premium pool</i> , never Jio revenue.	Premiums belong to the insurer, B2C uptake is a guess, and big numbers invite attack without adding proof.
"Later, workers add Kavach Basic at recharge."	Reliance first, then businesses. "Basic at recharge" stays only as an optional Year-2 roadmap item.	Selling insurance directly to individuals is hard (trust, adverse selection, licensing). B2B decisions are a clearer first market.

"Why only Reliance?... No other telecom has..."	"The edge is integration, not the sensor."	"Only" is absolute and easy to attack (Airtel has towers; weather firms have radar). Integration across Sense → Understand → Reach → Protect is defensible.
"It pays three ways" (premiums, recharge add-on, data sales).	"How Jio creates value": intelligence fees (external), savings inside Reliance (internal), distribution fees (optional).	Savings are not revenue, and the business must stand even if insurance never launches.
Heat in Nagpur and Ahmedabad as Phase 2 of the same tower system.	Rain from Jio's links; heat from IMD stations and satellite through the same engine.	Microwave attenuation is rain physics. Links do not sense temperature.
"About forty radars can't see that difference."	Radar is an input we combine, not a rival.	Attacking IMD invites "then just use radar". Our position: links + radar + gauges, and accuracy decides the mix.
Phase 1: 90-day Mumbai audit with shadow payouts.	Three gates (coverage, accuracy, value), overlapping phases; the coverage audit selects the live-test city; stop or scale.	Clear go/no-go points, and we cannot pre-select a city before knowing its coverage.

Fixes made while drafting the deck

EARLIER DRAFT	FINAL	WHY
"Reliance is the first customer"	"first proving ground"	Reliance has not agreed to buy anything.
Six internal uses (incl. JioAds, Reliance Foundation, JFS)	Three: network ops (hero), JioMart, MyJio	Fewer, provable uses beat "weather for everything".
"Layer the sources, cheapest first"	"Let accuracy determine the mix"	An insurance-grade system should not start from cheapest.
"A sensor on every link, far denser than gauges, early warning, strongest in rural areas, independent trigger"	"Dense sensing wherever usable links exist", "potentially denser", "near-real-time monitoring", "auditable multi-source trigger"	Each original claim was unproven or wrong (detection is not prediction; Jio's own system is not "independent").
"Others hold one or two of these layers"	Deleted	Unprovable competitor claim.
"Low cost", "Free" cost labels	Source labels: Jio-owned, IMD (govt), IMD · BMC, External, New install, Commercial	We cannot prove prices; we can state who provides each source.
"18% false triggers"	"18% of our triggers not confirmed by radar"	Radar is not perfect truth.
Heat cards at 45°C (9.7 and 5.4 days)	47°C (2.2 and 0.7 days), matching the price table	The cards and the product used different triggers.
"for platforms to sponsor"	"designed for potential platform sponsorship, subject to pilot validation"	Slide 8 says who pays is unproven.
"Indian weather sets the price"	"weather risk sets the starting price"	Loss ratio, costs and reinsurance also set price.

What we kept from the video

- The 944 mm vs 73 mm Mumbai story (26 July 2005)
- Mumbai first, rain first, heat as Phase 2
- About 12 heavy-rain days a year in Mumbai (IMD, 2000–2024)
- Kavach Basic ₹45 / ₹52 / ₹29 and Plus ₹208 a month
- 90 days with shadow payouts first; IRDAI sandbox and heat in months 4–12
- The roadmap labels (Basic at recharge in Year 2, national scale in Year 3)

IF THEY ASK “WHY IS THE DECK DIFFERENT FROM YOUR VIDEO?”

“The video was our first version. Afterwards we stress-tested three things: accuracy, who pays, and evidence. Our own test showed links alone catch 68% of rain events, not enough to pay claims on, and we couldn’t prove who would pay for insurance. So we kept the core, Jio’s network signal and the Mumbai problem, and moved insurance from the anchor to an option. The video’s 95% was decision agreement including dry days; we now lead with the stricter 68%.”

The final deck, slide by slide

For each slide: why it exists, what is on screen, what you say, where every number comes from, and the trap to avoid. Labels: **FACT** **OUR CALCULATION** **OUR TARGET** **OUR ASSUMPTION** **PLAN**.

SLIDE 1 India has a weather-decision problem

Why it's there: Make the problem concrete in ten seconds: the data exists, but it is too coarse to act on zone by zone.

ON SCREEN

- Bar chart: Santacruz 944 mm vs Colaba 73 mm, 26 July 2005, stations ~20 km apart (IMD)
- "What a city-level alert says: Heavy rain in Mumbai" → "What it needs to decide: Which zones do I act on, right now?"
- Businesses decide blind · Workers carry the loss

WHAT YOU SAY

Imagine you're running deliveries in Mumbai. Your dashboard says: "Heavy rain in Mumbai." Do you stop the whole city? On 26 July 2005, Santacruz got 944 millimetres of rain. Colaba, 20 kilometres away, got 73. The problem isn't weather data. It's knowing what to do, zone by zone.

NUMBERS AND WHERE THEY COME FROM

944 vs 73 mm

FACT

IMD rainfall records

Watch out: Present the alert as imagined ("Imagine you're running deliveries..."), not as something a named business receives. If they say "one day proves nothing", use the 2020–24 station comparison (Part C, "The problem and the idea").

SLIDE 2 Jio's microwave links may carry a weather signal

Why it's there: The insight, and a precise definition of JioMausam.

ON SCREEN

- Two towers and a microwave link; rain weakens the signal; measure the excess loss → path-averaged rainfall, near-real-time
- "This technique uses microwave links, not fibre."
- JioMausam is decision infrastructure, not a weather map: link signals + IMD radar + gauges → spatial weather grid → decision-ready alerts and triggers
- It is / It isn't (not a new sensor network, not a consumer weather app, not an insurer, no heat sensing from towers)
- Takeaway: "Jio may already own the infrastructure for a new weather-sensing layer."

WHAT YOU SAY

But Jio may already own infrastructure that can carry a signal about that rain. Rain weakens microwave signals between telecom sites. Measure the extra loss and you can estimate rain along each link. JioMausam combines that with radar and gauges into alerts businesses can act on.

Watch out: Keep "may". Never say "every tower is a rain gauge" or that Jio can already sense the weather.

SLIDE 3 Links alone are promising, but fall short of our bar

Why it's there: Credibility: show what is proven, what we checked ourselves, what is not proven, and the honest limit.

ON SCREEN

- Proven · published: the physics; Science 2006; PNAS 2013 (rain mapped across a whole country)
- Our check · public data: we ran the open-source method on 500 real links with radar as reference
- Not yet proven · the pilot: Jio's coverage, Indian accuracy, business value, who pays, insurance-grade triggers
- Banner: OUR CHECK · 500 REAL LINKS · PUBLIC RESEARCH DATA, GERMANY · NOT JIO DATA
- 68% of rain events caught by links alone, against our 80% bar; event = 10 mm+ in 3 hours in a ~4 km zone; 8 unseen test days
- Chart: network average, hour by hour, links vs radar

WHAT YOU SAY

Sounds great? We thought so too. The physics is well established, so we tested it ourselves: 500 real links from a public dataset in Germany, not Jio's. The links tracked radar closely, but alone they caught only 68% of rain events. Our bar is 80%. That gap told us how to build JioMausam.

NUMBERS AND WHERE THEY COME FROM

68%	OUR CALCULATION	pycomlink public dataset (KIT), German radar
80%	OUR TARGET	Pilot go/no-go bar
10 mm in 3 h	OUR ASSUMPTION	Event rule; 5 and 15 mm also tested
Science 2006, PNAS 2013	FACT	Published papers

Watch out: Never say "studies show 68%" (it is ours) or "we tested Jio's network". Mention 0.95 only if asked.

SLIDE 4 Let accuracy determine the mix

Why it's there: Answer the limit: combine sources, test every mix, keep the cheapest that clears the bar.

ON SCREEN

- Sources and providers: Jio microwave links (Jio-owned) · network health: faults, outages, maintenance (Jio-owned) · Doppler radar (IMD, govt) · rain gauges (IMD · BMC) · satellite (External) · new gauges on Jio sites (New install) · commercial data (Commercial)
- Mix bars scored against held-out rain gauges: Links only = our check 68%; other mixes “measured in the pilot”; “Radar + gauges, no links” shows what Jio adds
- Rule: the lowest-cost mix that clears every accuracy gate ($\geq 80\%$ of predefined trigger events, totals within $\pm 10\%$, under 15% unconfirmed)

WHAT YOU SAY

So we don't bet on one sensor. Accuracy decides the mix: Jio's links, network-health data to filter out faults, radar, gauges, satellite. The pilot tests every combination against held-out gauges. The answer isn't “Jio only”. It's the cheapest mix that works.

NUMBERS AND WHERE THEY COME FROM

68%	OUR CALCULATION	Same as slide 3
80%, $\pm 10\%$, <15%	OUR TARGETS	Pilot gates

Watch out: The pilot does the scoring; we have not done it yet. No prices are claimed; labels show who provides each source.

SLIDE 5 Reliance is the first proving ground

Why it's there: Remove the biggest commercial risk: prove value inside Reliance before selling outside.

ON SCREEN

- Hero: Jio network operations. Can we tell rain-related signal loss from equipment faults before sending a crew? Potentially fewer wasted visits and earlier flood alerts. Jio owns the network data and controls the operational decision.
- “Before Jio sells weather intelligence to anyone, it uses it to understand its own network.”
- JioMart and Reliance Retail: rain-aware dispatch, delivery promises, zone pauses (tests commercial value)
- MyJio: street-level rain alerts (tests reach at scale)

WHAT YOU SAY

Before selling to Swiggy, where do we prove it? Inside Reliance, starting with Jio. A signal drops: is it rain, or is something broken? If JioMausam can answer that before a crew goes out, that's real value. Then JioMart tests commercial value; MyJio tests reach.

Watch out: Everything here is a hypothesis to test, not a result. Say “proving ground”, never “first customer”.

SLIDE 6 Then sell the same intelligence to the market

Why it's there: External market, and what the product actually is.

ON SCREEN

- Tiers: 1 delivery, quick commerce, ride-hailing (Alerts) · 2 logistics, cities and transport, insurers (Alerts + data) · 3 agriculture, power, construction, retail (Data + alerts)
- Example chain: Rain hits Zone A → JioMausam sends zone alert + rain estimate → the customer decides (dispatch, ETAs, incentives) → lower disruption
- Why buy from us: dense sensing wherever usable links exist · potentially denser than gauges · near-real-time monitoring · potentially valuable wherever microwave backhaul is present · auditable trigger from multiple sources

WHAT YOU SAY

Once that works, the same intelligence goes outside. Rain hits Zone A. JioMausam flags it. The platform decides whether to reroute, delay or raise incentives. That's the product: not a prettier weather map, but a signal that changes a business decision.

Watch out: Company names are examples of buyers, not customers we have spoken to. Jio sells intelligence; the customer owns the decision.

SLIDE 7 The edge is integration, not the sensor

Why it's there: Why Jio, and how value is created without insurance risk.

ON SCREEN

- Sense (Jio network) → Understand (JioMausam) → Reach (Reliance ecosystem) → Protect (JFS + licensed insurers, optional)
- "Reliance can connect all four layers, so Jio earns from the intelligence without carrying insurance risk."
- How Jio creates value: O1 external revenue, intelligence fees · O2 external, optional, distribution fees · O3 internal value, savings inside Reliance

WHAT YOU SAY

So why Jio? Not because it invented the physics. The edge is integration: Jio's signal, JioMausam's intelligence, Reliance's proving ground and reach. We can monetise the intelligence outside and create savings inside Reliance, even if insurance never launches.

Watch out: No claims about competitors. JFS is a separately listed group company; any deal would be at arm's length.

SLIDE 8 Insurance is one application, not the anchor

Why it's there: Show the pivot from the video, and that insurance is gated.

ON SCREEN

- Flow: qualifying weather event in a zone → JioMausam supplies the trigger data → a licensed insurer would underwrite and pay → fixed benefit to the worker, e.g. ₹300
- Eligible = already active in the zone before the event began (predefined rules); ₹300 is a prototype assumption
- Why optional: only after accuracy and actuarial gates (shadow payouts until then) · who pays is unproven · the insurer carries the risk, never Jio · if it fails, nothing breaks

WHAT YOU SAY

Kavach is where we started: protection for workers. Then we stress-tested it. Who pays? Is the trigger accurate? Can an insurer price it? The insurance isn't the business; the intelligence underneath it is. So Kavach becomes one application, only after the accuracy and actuarial gates pass.

NUMBERS AND WHERE THEY COME FROM

₹300

OUR ASSUMPTION

Prototype benefit

Watch out: No insurer partner yet: always “would”. Kicker reads “Where we started”: own the pivot.

SLIDE 9 If we offer Kavach, weather risk sets the starting price

Why it's there: Keep continuity with the video's pricing, and show prices come from data.

ON SCREEN

- Table: Mumbai Basic ≥ 204.5 mm/day (1.1 days → ₹45) · Nagpur $\geq 47^\circ\text{C}$ (2.2 raw, ~1.2 paid → ₹52) · Ahmedabad $\geq 47^\circ\text{C}$ (0.7, ~0.6 → ₹29 floor) · Mumbai Plus ≥ 64.5 mm/day (11.9 raw, 5 paid → ₹208)
- Price = expected paid days $\times$ ₹300 $\div$ 60% $\div$ 12
- City cards: Mumbai 11.9 heavy-rain days a year (Phase 1) · Nagpur 2.2 and Ahmedabad 0.7 days at 47°C + (Phase 2)
- Footer: ₹300 and 60% are prototype assumptions; 5-day annual cap; final pricing subject to actuarial validation

WHAT YOU SAY

If we offer it, Indian weather risk sets the starting point for the price. From 25 years of IMD records: about 45 rupees a month in Mumbai, 52 in Nagpur, 29 in Ahmedabad, and 208 for Mumbai Plus. All illustrative until actuarial validation.

NUMBERS AND WHERE THEY COME FROM

Trigger levels 204.5 / 64.5 mm

FACT

IMD rain categories

47°C

OUR ASSUMPTION

Rare heat level

Days a year, paid days

OUR CALCULATION

Counted from 25 years of IMD station reports

Prices

OUR CALCULATION

Formula above

Watch out: ₹208 is not priced from the 10 mm-in-3-hours rule. All prices are illustrative.

SLIDE 10 Don't believe us. Test it in 90 days

Why it's there: A concrete, low-risk ask with stop/scale points.

ON SCREEN

- Days 1–30 Coverage (Gate 1: enough usable coverage?) · Days 15–60 Accuracy (Gate 2: fused system clears every gate?) · Days 61–90 Value (Gate 3: improves a real Reliance operation? matched zones with and without JioMausam)
- Roadmap: 0–3 m three gates / shadow payouts · 4–12 m Reliance live + first clients / sandbox + heat · Year 2 cities + enterprises / Basic at recharge · Year 3 national weather API / national scale
- “Only after all three gates pass do we scale externally.”
- Close: “We aren’t asking Jio to build another weather network...”; “First prove the signal. Then sell the intelligence. Protection comes last.”

WHAT YOU SAY

We're not asking you to believe us. We're asking Jio to test it. Ninety days, three questions. Does Jio have the coverage? Does the combined system clear our bar? Does it improve a real Reliance operation? If yes, scale. If not, stop. First prove the signal. Then sell the intelligence. Protection comes last.

NUMBERS AND WHERE THEY COME FROM

90 days, gates

PLAN / TARGETS

Our pilot design

Watch out: “We don’t pre-select the city; the coverage audit does.”

Question bank

Answer in two or three sentences, then stop. Every answer should end by pointing back to the strategy: combine sources, prove it inside Reliance, gates before scale.

The 12 questions to know cold

Q What is JioMausam, in one line?

A weather-intelligence layer built from signals Jio's microwave network may already carry, combined with radar and gauges and turned into zone-level alerts that businesses act on. Insurance is one optional application on top.

Q Are you claiming Jio already has a working weather network?

No. We know the physics works, and our own links-alone test shows it isn't enough on its own. We don't yet know Jio's usable coverage, Indian performance or commercial value. So we're asking for 90 days to measure those, with explicit gates and a stop-or-scale decision.

Q How did you get 68%?

A public dataset from KIT in Germany: 500 real phone-network links, signal every minute for 11 days, with German weather radar as the answer key. Rain weakens the signal; a standard formula turns that loss into rain. For every ~4 km zone each test day we asked: 10 mm in 3 hours? Radar said yes 1,639 times; links also said yes on 1,120. $1,120 \div 1,639 = 68\%$. Below our 80% bar, which is why we never use links alone.

Q Why 80%?

Our go/no-go bar for the pilot, set deliberately above the 68% links alone achieved, so passing proves the combined system adds value. It sits with two other gates so you can't pass by triggering every day, and an insurer or use case can tighten it.

Q Why would Swiggy or Zomato pay?

We don't assume they will; that's why we prove value inside Reliance first. The value case: rain shifts demand and rider supply zone by zone, so better zone decisions mean fewer late orders, less wasted incentive spend and fewer cancellations. Gate 3 measures it with matched zones before we sell outside.

Q Why can't Google or a weather company do this?

They forecast and they sell weather data, so they're potential inputs or partners. What they don't have is the signal inside Jio's network, or Reliance operations to prove it on. Our edge is integration, not the sensor.

Q What stops Airtel from copying you?

Nothing in the physics; the method is published. The edge is speed and integration: Jio's network data plus Reliance's operations, reach and financial ecosystem. If Airtel does it too, that validates the market, and businesses will buy whichever is more accurate in their zones.

Q What if Jio doesn't have enough microwave links?

Then Gate 1 tells us in the first month, cheaply. The system still runs on radar, gauges and network-health data for operations, and we can add gauges on Jio sites where links are thin. The question becomes how much Jio's links add, not whether the business exists.

Q Why include insurance at all?

It's where we started and it matters to workers. But it launches only after the accuracy and actuarial gates pass, with a licensed insurer carrying the risk. If it never launches, the intelligence business still stands.

Q How did you price Kavach?

From 25 years of IMD station records we counted how often each trigger happens, capped paid days at 5 a year, and applied: $\text{paid days} \times ₹300 \div 60\% \div 12$. ₹300 and 60% are prototype assumptions; an actuary sets final prices.

Q What do you need from Jio for the pilot?

Mostly access, not money: link telemetry and network-health logs for one city, a small data team with a meteorology partner, IMD/BMC data, and one Reliance operation willing to run a shadow test. We'd cost it properly with Jio once Gate 1 shows what data exists.

Q What is your single biggest risk?

Coverage: whether Jio has enough usable microwave links where it matters. That's exactly why it is Gate 1, and it costs almost nothing to check.

Competition and alternatives

Judges will test whether someone bigger already does this. Never attack competitors; position them as inputs, partners or proof the market is real.

Q Google and DeepMind already do AI weather forecasting. Why you?

Forecasting and observing are different jobs. AI forecasting models predict weather hours to days ahead from global data. JioMausam observes rain happening now, zone by zone, and turns it into operational alerts. They complement each other: a forecast says rain is likely this afternoon; JioMausam says it is raining in Zone A now. A forecast model could even use our observations as an input.

Q Weather companies already sell hyperlocal weather APIs (e.g. The Weather Company, Tomorrow.io, Skymet).

Yes, and platforms already buy from them. Their hyperlocal products are mostly models built on radar, satellite and stations. None of them has Jio's network signal. If our links add accuracy, that's a real edge; if they don't, our pilot's "radar + gauges, no links" baseline shows it and we won't pretend otherwise. Some of these firms could be partners or data buyers.

Q Hasn't this been tried commercially already?

The idea isn't new, which is what makes it credible. Researchers proved it in Israel (2006) and mapped a whole country in the Netherlands (2013), and weather startups and telecom vendors have built or piloted products on wireless-signal data. What's new is doing it in India, on Jio's network, combined with IMD data and proven inside Reliance operations.

If pushed: Researchers and companies have worked on this for over a decade. We'd rather show our own check than quote other people's claims.

Q IMD has radars and is expanding under Mission Mausam. Doesn't that make you redundant?

No. IMD is an input and a natural partner. Its radars and gauges are in our mix. IMD's job is public forecasts and warnings; ours is turning observations into decisions for specific operations. More IMD radars make JioMausam better, not obsolete.

Q Mumbai already has BMC rain gauges and hyperlocal weather apps.

Exactly, and they're inputs we combine, not rivals. The real question is whether Jio's links add information between gauges. The pilot measures that directly with the "radar + gauges, no links" run.

Q What stops Airtel, Vi or BSNL?

Nothing technical; the method is open. The edge is integration and speed: Jio's network signal plus Reliance's operations to prove value on, its enterprise reach and its financial ecosystem. A competitor doing it would confirm the market.

If pushed: "Could you work with other operators?" Possibly later: more links mean better coverage. We'd start with Jio's own network.

Q Why not tower companies? They own the towers.

They own the steel and the sites. The radio links and their signal data belong to the operator running them. That's Jio.

Q Delivery platforms can build this in-house.

They can and should build their own decision logic. What they can't build is the raw network signal. Our product is the intelligence feed they plug into their own dispatch systems.

Q Why not just buy weather data and skip Jio's links?

That is our pilot baseline. If links add nothing, we say so; then the case rests on the decision layer and Reliance's distribution, and the moat is weaker. Better to learn that in 90 days than after a big launch.

Q Insurers already sell parametric weather insurance (e.g. crop insurance).

Yes: India already runs a weather-index crop insurance scheme (WBCIS), so parametric cover is familiar to insurers and the regulator. That's why Kavach plugs into licensed insurers instead of competing with them. We supply trigger data.

Q Satellites cover all of India. Why not use them?

Satellite rainfall is open and covers everything, but it's coarse and can lag, so it's our fallback, not the main source. The pilot evaluates satellite sources on coverage, latency, resolution and licensing.

If pushed: Open satellite rainfall products are roughly 10 km squares, far coarser than zone decisions need.

Q Why not put IoT rain sensors everywhere?

Hardware at scale means capex, maintenance and theft. We add gauges only where links are thin, on Jio sites that already have power, security and connectivity.

Q Why not crowdsource from smartphones?

Phones don't measure rain; user reports are noisy and sparse in heavy rain. It could become a confirmation signal later, but it isn't the core.

Q Why not use 5G or phone-to-tower signals?

Backhaul links are fixed point-to-point paths with known length and frequency, so the physics is clean. Phone-to-tower signals vary with movement, buildings and indoor use, so rain is much harder to separate out.

The problem and the idea

Show the problem is real without overclaiming.

Q Is this a real problem? Platforms already handle rain.

They handle it with city-level information and manual calls. Platforms charging a rain fee shows rain clearly matters to their economics. Our claim is narrow: zone-level precision could improve those decisions, and Gate 3 measures whether it does.

Q One day in 2005 isn't evidence.

Agreed, so we checked recent years too. In the 2020–2024 monsoons, Santacruz and Colaba both reported on 89 days. On 15 of them at least one station had heavy rain (64.5 mm+), and on 5 of those 15 the other station got less than half. Rain differing sharply across one city is routine, not a one-off.

❓ Isn't this a solution looking for a problem?

We started from a problem, riders losing income and businesses deciding blind in rain, and stress-tested our way to the solution. And we test value before scaling, so if the problem isn't worth solving, Gate 3 says so.

❓ Why rain and not floods or air quality?

Rain is what microwave physics measures. Flood alerts for sites are a downstream use of rain data. Air quality can't be measured this way.

❓ Why Mumbai?

Most heavy-rain days of our three cities (about 12 a year), dense operations, and the clearest example of rain varying within a city. But the live-test city is chosen by the coverage audit, not by us.

❓ "Decision infrastructure" sounds like jargon.

It means we don't sell a weather map. We sell alerts and triggers that plug into a business's own decisions: dispatch, pausing routes, incentives, payouts.

The technology

Plain answers; admit what the pilot must measure.

❓ How exactly does rain weaken the signal?

Raindrops absorb and scatter microwave energy. The heavier the rain, the bigger the loss. The relationship is an international standard (ITU-R P.838) and depends on the link's frequency, polarization and length.

❓ Which frequencies work?

Links in our test ran from about 6 to 39 GHz; higher frequencies react more strongly to rain. Very short, very high-frequency links are highly sensitive but can drop out in extreme rain. The coverage audit lists which of Jio's bands are usable.

❓ Jio is mostly fibre. Doesn't that kill the idea?

In dense cities fibre may dominate; fibre gives no rain signal. That's why coverage is Gate 1. Where links are thin we lean on radar, gauges and new gauges on Jio sites. The business doesn't depend on Jio's links alone; the pilot measures what they add.

❓ In very heavy rain, doesn't the link just drop?

Some links can, at extreme rain. A drop-out itself tells you rain passed a high level, and radar and gauges cover the gap. We treat outages as information, not missing data.

❓ How do you tell rain from an equipment fault?

Three ways: the pattern (rain builds and fades; faults are sudden), neighbours (rain hits nearby links together; a fault hits one), and Jio's network-health flags for faults, outages and maintenance. That's why network health is a source on slide 4.

❓ What is the wet-antenna problem?

Water on the antenna adds loss that isn't rain along the path. We correct it with a standard method (Schleiss et al., 2013), allowing up to 1.5 dB.

Q What about fog, humidity and temperature?

Their effect is much smaller than rain at these frequencies and changes slowly, so the dry-weather baseline absorbs it.

Q Does Jio even store this data? How often?

Operators' network management systems record signal levels; often as 15-minute minimum and maximum values. The Netherlands country-wide maps were built from exactly that kind of operator data. How often and how far back Jio's data goes is part of Gate 1. Live minute-level data would need polling.

Q A link averages rain along its path. How do you know where it rained?

Right: one link gives an average. Many crossing links plus radar and gauges locate the rain; that's the fusion step that builds the grid.

Q What resolution will you get?

It depends on link density and layout. Our German check used ~4 km zones. We don't promise 1 km everywhere; the pilot finds the real resolution city by city.

Q How fast are alerts?

Near-real-time; how near depends on how fast telemetry can be pulled. Operational decisions need minutes, not seconds. The pilot measures it.

Q Can JioMausam forecast?

It observes current rain. Short-term forecasts could be built on top later by tracking movement, but we don't claim that.

Q Is the ITU formula valid for Indian monsoon rain?

The formula assumes typical raindrop sizes; monsoon rain can differ. That's exactly why local calibration against Indian gauges is part of Gate 2.

Q How much data and computing does this need?

Modest: a few numbers per link per minute. Jio already processes far larger telemetry every day.

Q Will Jio's security team allow this?

Data stays inside Jio. JioMausam only outputs aggregated rain per zone, never network layout or customer data. Network performance readings aren't personal data.

Our 68% check

Know this cold. Answer, then return to strategy.

Q Why German data and not Indian or Jio data?

Jio's data isn't public. This is the standard public research dataset with real links and a radar answer key. Jio's real number is exactly what the pilot measures.

Q Only 11 days?

Yes: it's a check of the method, not a final validation. It still gives 14,256 yes/no decisions on 8 unseen days. The pilot uses a full monsoon.

Q May in Germany isn't a monsoon.

Agreed. Monsoon rain is heavier and longer; heavier rain gives a stronger signal but more drop-out risk. That's why Gate 2 back-tests archived Mumbai monsoon data.

Q Radar isn't ground truth.

Agreed; it was the reference in that dataset. That's why we say "not confirmed by radar", not "false". The pilot checks against held-out rain gauges.

Q Did you tune it to look good?

We tuned one scale factor on the first 3 days only, then tested on the 8 days it had never seen.

Q One scale factor for 500 links sounds crude.

It is, deliberately. Production systems calibrate locally. Our result is a floor, not a ceiling.

Q How closely did links track radar?

Network-wide hourly match 0.95; a typical single link 0.88; map level 0.78. The trigger test is harsher: a 9 mm estimate against an 11 mm event counts as a miss.

Q Why do links read 18% low?

The tuning days had very little rain, so the scale factor was fitted on light rain, and path-averaging smooths peaks. Local calibration in the pilot addresses it.

Q Why 10 mm in 3 hours?

Our heavy-shower rule for the check. We also tried 5 mm and 15 mm: 64% and 54% caught. None got near 80%, so the conclusion doesn't depend on the rule.

Q Which software?

Python with pycomlink, the open-source toolbox from KIT that implements the published method, plus our own code for the zone map and the comparison.

Q Can we check your work?

Yes. The data is public, the code is in our repository, the workbook recomputes 68% from the 14,256 decisions, and our Test Lab reruns everything in a browser.

Q Did you build a model?

No new model and no machine learning. We ran the published method on public data and calculated the result ourselves.

Q Is 68% good or bad?

For one source with simple processing it's a reasonable floor. For paying claims it isn't enough. That gap is why the product combines sources.

Q What is the 18% unconfirmed?

Of our 1,364 triggers, 244 weren't seen by radar. Some are our errors, some may be radar misses.

Accuracy gates and combining sources

Q Why $\pm 10\%$ on rain totals?

If amounts are off, prices and payouts are off. Our links-alone check read 18% low, so this gate forces calibration.

Q Why under 15% unconfirmed?

So a system can't "pass" the 80% gate by triggering every day. Our check had 18%.

Q Who sets the gates?

We set them as the pilot's go/no-go. Jio, and for Kavach an insurer, can tighten them.

Q What are held-out gauges?

Gauges kept hidden from the system while it's tuned, and used only to score it, so we're not marking our own homework.

Q What if no combination clears 80%?

Then we don't scale externally and Kavach stays in shadow or is dropped. We'd keep only uses that need less precision, like network operations, if Gate 3 shows value.

Q How do you know what each source costs?

We don't claim prices. The slide shows who provides each source. The pilot puts a rupee figure on each, and the rule picks the cheapest mix that clears the bar.

Q What is "network health" data?

Jio's own alarms and logs: equipment faults, outages, maintenance windows. It removes signal drops that aren't rain.

Q What would new gauges on Jio sites cost?

Priced in the pilot. They're only for areas where links are thin, and Jio sites already have power, security and connectivity.

Q When would you buy commercial data?

Only if it lifts accuracy enough to justify its cost. Otherwise not.

Reliance as the proving ground

Q Why start inside Reliance?

Jio owns the data and the decisions, savings are measurable, and there's no sales cycle. It proves value before anyone outside has to pay.

Q Doesn't Jio's network team already know rain fade?

It may. Then Gate 1 confirms cheaply what weather information is available. If it doesn't, that's new value. Either way the same telemetry feeds the weather layer.

Q How would network operations save money?

By not sending crews for rain-related signal loss that recovers on its own, prioritising real faults, and warning sites at flood risk. Gate 3 measures site visits avoided, fault-resolution time and weather outages correctly identified.

Q What does JioMart get?

Rain-aware delivery promises, dispatch and zone pauses. Measured by late deliveries, cancellations and rider availability.

Q You said it isn't a consumer app, but MyJio sends alerts.

MyJio tests reach inside an existing app. We're not building a standalone consumer weather app business.

Q Will Reliance businesses actually cooperate?

That's part of the ask: one operation for one shadow test. The coverage audit decides which city.

Market, customers and selling

Q What exactly do you sell?

Zone alerts (rain starting, intensity), live API feeds, historical rain records for audits and claims, and trigger data for insurers.

Q How will you price the intelligence product?

After the pilot, benchmarked to the value Gate 3 measures. Likely a subscription per city or zone, or by API volume. We won't guess numbers before we have evidence.

Q What is the market size?

We deliberately don't headline one; we'd build it bottom-up after the pilot. For Kavach alone, 50,000 Mumbai riders on Plus is about a ₹12.5 crore premium pool a year, which is the insurer's money, not Jio revenue.

If pushed: If asked about the video's ₹174 crore: $50 \text{ crore users} \times 1\% \times ₹29 \times 12$. It is an illustrative premium pool, B2C uptake is unproven, so we removed it.

Q Why delivery first?

It makes the most rain-sensitive real-time decisions, zone by zone, many times a day.

Q Why would BMC or a city buy?

Flood response, pump deployment, traffic management. Public-sector sales are slower, so cities are tier 2.

Q What about agriculture?

Rainfall records for crop insurance and advisories; rural areas may rely more on microwave backhaul. Tier 3, and unproven.

Q How do you reach customers?

Potentially through Jio's enterprise sales relationships and Reliance's partners. That's the "Reach" layer.

Q Why buy from Jio rather than a weather firm?

Only if Jio's links add accuracy in the buyer's zones, plus an auditable multi-source trigger and integration with Reliance's ecosystem. If links don't add accuracy, we don't claim an edge.

Why Jio, the moat and the economics

Q The method is open-source. Where is the moat?

Access to Jio's private link data, calibration history built over seasons, integration with Reliance operations, and distribution. The method is open; the signal isn't.

Q Why should Reliance invest?

Low capex because the infrastructure exists, internal savings first, optional upside, and a strategic data asset. And it costs 90 days to find out.

Q Where are your revenue projections?

None on the slide on purpose. We'd rather show value measured in Gate 3 than invented forecasts.

Q Savings inside Reliance aren't revenue.

Correct, which is why slide 7 separates external revenue from internal value.

Q JFS isn't part of Reliance any more.

It's a separately listed group company and a natural partner. Any arrangement would be at arm's length, with a licensed insurer carrying the risk.

Q What does Jio actually risk?

A small pilot. No insurance risk; shadow mode protects reputation.

Q How big are distribution fees?

Illustratively, 15–25% of a ₹12.5 crore premium pool is ₹1.9–3.1 crore. Small, which is exactly why insurance is not the business.

Kavach (insurance)

Keep Kavach clearly optional and clearly the insurer's risk.

Q What is parametric insurance?

It pays a fixed amount when a measured trigger is crossed, like rain above a level in your zone, without assessing individual losses.

Q Why can't Jio be the insurer?

Insurance needs an IRDAI licence and capital. Jio supplies data; a licensed insurer underwrites and pays.

Q What is basis risk?

A worker gets rained on but no payout triggers, or the reverse. Small zones, multi-source triggers and a review process reduce it, which is why the accuracy gates come first.

Q Won't people log in just to collect payouts?

Eligibility requires being already active in the zone before the event began, checked against platform activity logs.

Q Won't people buy only when heavy rain is forecast?

That's adverse selection. The answer is season-long or annual cover, or platform group cover, with no buying in mid-event.

Q Who pays: platforms or riders?

Unproven: platform-sponsored, co-funded or individual. The pilot tests it.

Q Why would a platform pay ₹2,500 a year for at most ₹1,500 of payouts?

The gap is the insurer's costs at an assumed 60% loss ratio. A platform could fund the allowance itself using our trigger data, and Jio earns from the data either way.

Q How is the money paid? JioPay?

Through the insurer's payment rail (UPI or bank). We don't depend on any particular wallet. Shadow mode measures the speed.

Q What is the IRDAI sandbox and shadow mode?

Shadow mode computes would-be payouts without moving money. The regulator's sandbox lets insurers test new products within limits. Real payouts come only after the gates.

Q Why ₹300?

A round prototype figure, roughly part of a day's earnings. The insurer or platform would set the real benefit.

Pricing

Every price is illustrative. Know the formula and one worked example.

Q Walk me through ₹45.

Mumbai sees ≥ 204.5 mm in a day about 1.08 times a year. $1.08 \times ₹300 = ₹324$ of expected payouts; $\div 0.6 = ₹540$ a year; $\div 12 = ₹45$ a month.

Q And ₹52, ₹29, ₹208?

Nagpur: 2.2 days a year at $47^\circ\text{C}+$, but capped at 5 paid days a year that averages 1.24 paid days $\rightarrow ₹52$.

Ahmedabad: 0.64 paid days $\rightarrow ₹27$, raised to our ₹29 minimum price. Mumbai Plus: ≥ 64.5 mm happens about 12 days a year, every year at least 5, so it always pays 5: $5 \times 300 \div 0.6 \div 12 = ₹208$.

Q Why a 60% loss ratio?

An illustrative assumption: ₹60 of every ₹100 goes back as payouts, ₹40 covers costs and margin. An actuary sets the real figure.

Q Why cap at 5 days a year?

It limits the worst-year cost and keeps the price predictable. With a 3-day cap, Plus would be ₹125.

Q Why ₹29 for Ahmedabad when the maths says ₹27?

Our chosen minimum price, so even very low-risk cover pays for basic administration.

Q Why 47°C and not 45°C ?

45°C happens about 10 days a year in Nagpur, which would price Basic at about ₹168 a month. 47°C targets rare extremes at an affordable ₹52.

Q Why ≥ 204.5 mm for Mumbai Basic?

That's IMD's "extremely heavy rain" level: rare (about once a year), so Basic stays cheap. IMD's "heavy rain" level (64.5 mm) is the broader Plus product. A very-heavy trigger (115.6 mm) would price at ₹165.

Q Your live trigger is 10 mm in 3 hours. Why price on 64.5 mm a day?

₹208 isn't priced from that prototype rule. It uses 25 years of daily IMD heavy-rain history, capped at five paid days. The pilot measures the 3-hour trigger's real frequency and recalibrates before any launch.

Q What happens in bad years?

We replayed every year from 2000 to 2024. Worst years: Mumbai Basic 2019 paid 222% of premiums, Nagpur 2013 242%, Ahmedabad 2018 469%. Plus was 60% every year because it always hits its cap. Bad years are why a licensed insurer pools risk across cities and years, with reinsurance.

Q Is NOAA's data reliable?

It's NOAA's public archive of IMD's own station reports. We kept only years with at least 330 days of reports, including 110 in June–September; all 25 years passed. Missing days count as no trigger, so if anything we slightly undercount.

Q Why these three cities?

Mumbai is India's rain extreme; Nagpur and Ahmedabad are heat extremes. Together they show one flat price can't fit India.

Q Climate change makes 25-year averages wrong.

Trends matter; studies report a sharp rise in widespread extreme rain over central India since 1950. An actuary would weight recent years and add a loading. Our numbers are a starting point.

Q Would a rider pay ₹45 a month for about one payout a year?

Maybe not, which is why Basic is optional and who pays is tested. Basic is cheap cover for rare extremes, not an income product.

Pilot and roadmap

Q Is 90 days realistic?

Yes, because nothing new is built first. The audit is desk work on existing telemetry, the back-test uses archived data, and the live test is about a month. The phases overlap.

Q Which city, and when?

The coverage audit chooses. A live test now would run in a north-east-monsoon city (rain from October to December); Mumbai's 2027 monsoon falls in months 4–12.

Q What if it doesn't rain in days 61–90?

The back-test replays archived monsoon events, including network-operations value. The full live test runs in Mumbai's 2027 monsoon.

Q What are matched zones?

Zones using JioMausam compared with similar zones that aren't, on the same days. The difference is the value.

Q What are your stop criteria?

Any gate fails: stop, or narrow to what passed. No external scaling until all three pass.

Q Year 2 says "Basic at recharge". Didn't you say B2C is hard?

It's optional and only happens if Kavach passes its gates and an insurer partners. Recharge is just a distribution channel.

Q How do you get to a national API by Year 3?

City by city: audit coverage, calibrate locally, launch where coverage supports it.

Feasibility and scalability

Q Is it technically feasible?

The physics is proven, the tools are open, the data exists in operator systems, and we ran the method ourselves on real link data.

Q Operationally?

A small team on existing telemetry; no new hardware except optional gauges where links are thin.

Q Financially?

Low upfront cost, internal savings first, and a 90-day decision point before any big spend.

Q Does it scale?

The software scales with the network. Each new city needs a coverage audit and local calibration, and different climates need their own calibration.

Q What skills does it need?

Data engineering, signal processing and a meteorology partner, for example IMD or an academic institute.

Q Networks change. Links get replaced by fibre.

Yes, coverage evolves, so we re-audit. Because we combine sources, quality degrades gracefully instead of failing.

The video vs the deck

Judges saw the video. Own the change; don't hide it.

Q Your video said 95%. Now you say 68%. Which is it?

Both are true, but they measure different things. The 95% was decision agreement including dry days, when both systems say "no". The 68% is the stricter question: how many real rain events were caught. After stress-testing we lead with the stricter one.

Q Your video said every tower is a rain gauge.

We learned only microwave links work, not fibre, and nobody knows Jio's usable coverage yet. That's why coverage is now Gate 1.

Q Your video said ₹174 crore.

That was an illustrative premium pool from a B2C assumption we can't prove, and premiums aren't Jio revenue. We removed it.

Q Your video said ₹300 lands in a JioPay wallet within hours.

A licensed insurer would pay through its own rail. ₹300 is a prototype assumption, and nothing depends on JioPay.

Q Your video was riders first. Now it's Reliance first?

Riders still matter; Kavach is still on the roadmap. Proving value inside Reliance first lowers the risk before platforms are asked to pay.

Q Why should we trust a pitch that changed?

Because it changed for evidence. We tested our own idea and kept what survived.

Curveballs

🕒 Why now?

Quick commerce makes zone-level rain decisions worth more; extreme rain events are rising; operators keep adding microwave backhaul; and IMD keeps adding radars, which are inputs we combine.

🕒 If Reliance says no, what happens?

Then the 90-day audit is the cheapest way to say no with evidence instead of opinion.

🕒 If you had to cut one thing, what?

Kavach. The intelligence business stands without it.

🕒 What did you learn doing this?

That a flattering number (0.95) can hide the number that matters (68%), and that testing your own idea is the fastest way to improve it.

🕒 Explain it to a 10-year-old.

When it rains, the invisible radio beams between phone towers get weaker. We measure how much weaker to work out where it's raining, and tell businesses which areas to act on.

🕒 How is this "Different by Design"?

It uses infrastructure Jio already runs as a sensor, it's designed around decisions instead of maps, and it's built to be tested and stopped cheaply instead of launched on faith.

Backup plans: if an assumption fails

Every main assumption has a test, a Plan B and a Plan C. The design principle: combining sources means JioMausam gets weaker gracefully instead of failing outright, and every gate is a cheap place to stop.

ASSUMPTION	HOW WE FIND OUT	IF IT FAILS: PLAN B	PLAN C
Jio has enough usable microwave links where it matters	Gate 1 coverage audit (days 1–30): link counts, frequencies, path lengths, telemetry	Run the layer on radar + gauges + satellite, plus network-health data for Jio's own operations; focus first on areas where microwave is dense	Add rain gauges on Jio sites where links are thin; if coverage is poor everywhere, the Jio-specific edge is small, so stop external scaling
Jio's telemetry is stored often enough and far enough back	Gate 1: sampling interval, history kept, access route	Use 15-minute minimum/maximum data (it worked country-wide in the Netherlands); start live polling for forward collection	Use link alarm and outage logs as a coarse rain signal; back-test on whatever history exists
Combining sources reaches $\geq 80\%$ of trigger events, $\pm 10\%$ totals, $< 15\%$ unconfirmed	Gate 2 back-test on archived Mumbai monsoon data against held-out gauges	Local calibration, more gauges, better fusion; use the system for decisions that need less precision (network ops, soft alerts)	Kavach stays in shadow or is dropped; no external scaling
Jio's links add accuracy beyond radar + gauges	Gate 2: "radar + gauges, no links" baseline vs every mix with links	Keep links only where they add value (gaps between gauges, rural backhaul); sell the decision layer on the best mix	If links add nothing anywhere, say so: the moat shrinks to integration and distribution, and the go/no-go is decided on that basis
A Reliance operation sees measurable value	Gate 3: matched zones with vs without JioMausam (site visits, fault time, delays, cancellations, incentive spend)	Try another internal operation or a different decision (e.g. flood alerts for sites, staffing)	Stop. A cheap, evidence-based no
External businesses will pay	After Gate 3: paid pilots with one platform and one logistics firm	Bundle alerts into Jio's enterprise connectivity offers as a value-add	Keep it internal, or license data to insurers and agriculture
A licensed insurer partners for Kavach	Months 4–12: shadow results shared with insurers; regulator's sandbox	Platforms self-fund a weather allowance using our trigger data (no insurance needed)	Drop Kavach; the intelligence business is untouched
The price assumptions hold (₹300, 60%, 5-day cap)	Actuarial review; shadow-mode payout data	Change the benefit, cap or trigger (e.g. a 3-day cap brings Plus to ₹125)	Platform group cover instead of individual sales
Who pays: platforms, co-funding or individuals	Pilot willingness-to-pay tests with platforms and riders	Co-funded model (platform + rider)	Data-only product for platforms' own schemes
Security and legal teams allow the data use	Early review with Jio's security and legal teams	Process everything inside Jio; release only aggregated rain per zone	Use the layer internally only (network ops)

Accuracy holds in extreme rain (links can drop out)	Gate 2 back-test on the heaviest archived events	Treat drop-outs as “rain above a level”; lean on radar and gauges in those minutes	Flag extreme events for manual review before any payout
Heat can be covered by the same engine	Months 4–12: April–June heat season with IMD stations and satellite	Heat alerts first, insurance later	Leave heat out; rain only
Competitors (another telecom or a weather firm) move first	Market watch during the pilot	Compete on accuracy in our zones and on Reliance integration	Partner or license; a competitor confirms the market

How the system degrades

SITUATION	WHAT STILL RUNS
Full system	Jio links + network health + radar + gauges (+ satellite, new gauges, paid data only if they earn their cost)
Links thin	Radar + gauges + satellite, network health for Jio operations, gauges on Jio sites
Accuracy short	Soft operational alerts and network operations only; no payouts
No value shown	Stop, with evidence, after about 90 days

Stop criteria

- Gate 1: too few usable links and no telemetry access in the candidate cities → stop the Jio-links track; decide whether a radar-and-gauge decision layer is still worth it.
- Gate 2: no mix clears the accuracy gates → no external scaling; Kavach stays in shadow or is dropped.
- Gate 3: no measurable improvement in a Reliance operation → stop before spending on sales.
- Kavach: no insurer or no payer after shadow mode → drop Kavach; keep intelligence.

IF THEY ASK “WHAT IF YOUR MAIN ASSUMPTION IS WRONG?”

“Then we find out in the first 30 days, cheaply. If Jio’s links are thin, the layer still runs on radar and gauges for Jio’s own operations. If accuracy falls short, we don’t scale and insurance never launches. Every gate is a place to stop with evidence.”

Number guide, never-say list, cheat sheet

Every number and where it comes from

NUMBER	LABEL	SOURCE
944 vs 73 mm	FACT	IMD rainfall, Santacruz vs Colaba, 26 July 2005
15 heavy days, 5 with less than half	OUR CALCULATION	IMD station reports (NOAA archive), 2020–2024 monsoons
Science 2006 · PNAS 2013	FACT	Messer et al. (doi:10.1126/science.1120034) · Overeem et al. (doi:10.1073/pnas.1217961110)
500 links, 11 days	FACT	pycomlink public dataset (KIT), github.com/pycomlink/pycomlink
68% (1,120 of 1,639)	OUR CALCULATION	Our code on that dataset, German Weather Service radar as reference
0.95 · 0.88 · 0.78 · 18% low · 18% unconfirmed	OUR CALCULATION	Same run; Q&A only
80% · ±10% · <15%	OUR TARGET	Pilot gates
10 mm in 3 h, ~4 km zone	OUR ASSUMPTION	Event rule; 5 and 15 mm also tested (64%, 54%)
1.1 / 2.2 / 0.7 / 11.9 days a year	OUR CALCULATION	Counted from IMD station reports, 2000–2024
204.5 / 115.6 / 64.5 mm a day	FACT	IMD rain categories
47°C	OUR ASSUMPTION	Rare-heat trigger (45°C ≈ IMD heatwave level, too frequent)
₹45 / ₹52 / ₹29 / ₹208	OUR CALCULATION	Paid days × ₹300 ÷ 0.6 ÷ 12; ₹29 is our minimum
₹300 · 60% · 5-day cap	OUR ASSUMPTION	Prototype; actuary sets
222% / 242% / 469%; Plus 60%	OUR CALCULATION	Stress test, every year 2000–2024
₹12.5 crore pool; ₹1.9–3.1 crore fees	OUR CALCULATION	50,000 riders × ₹208 × 12; 15–25% fee (illustrative, Q&A only)
90 days, three gates	PLAN	Our pilot design

DATA LINKS

Rain-sensing test: github.com/pycomlink/pycomlink (folder `pycomlink/io/example_data`; also `pip install pycomlink`). Radar reference inside it: German Weather Service RADOLAN YW (opendata.dwd.de). India: noaa-gsod-pds.s3.amazonaws.com/{year}/{station}.csv, stations 43003099999 Santacruz, 43057099999 Colaba, 42867099999 Nagpur, 42647099999 Ahmedabad. Rechecks: the Numbers Workbook and the Test Lab in our repository.

Never say → say instead

NEVER SAY	SAY INSTEAD
"We tested Jio's network."	"We ran the published method on public research data from Germany."
"Studies show 68%."	"In our own check, links alone caught 68%."
"Jio can already sense the weather." / "Every tower is a rain gauge."	"Jio's microwave links may carry a usable weather signal."

"80% is the industry standard."	"80% is our go/no-go bar, set above the 68% baseline."
"Reliance is our first customer."	"Reliance is our first proving ground."
"₹300 goes to the rider's JioPay wallet."	"A licensed insurer would pay, through its own rail."
"We make ₹174 crore."	Don't raise it. If asked: an illustrative premium pool, not Jio revenue.
"Only Reliance can do this."	"The edge is integration, not the sensor."
"Low cost" / "free" about any source	"The pilot prices each source."
"We built a model."	"We ran the published method and calculated the result ourselves."
"False triggers."	"Triggers not confirmed by radar."
Naming a city for the live test	"The coverage audit selects the city."

The night-before cheat sheet

- 1 India doesn't have a weather-data problem; it has a weather-decision problem.
- 2 Jio's microwave links may carry a weather signal. Fibre doesn't.
- 3 The physics is well established. We tested it ourselves on public German data, not Jio's.
- 4 Links alone caught 68% of rain events (1,120 of 1,639). Our bar is 80%.
- 5 So accuracy decides the mix: links, network health, radar, gauges, satellite. Cheapest mix that clears the bar.
- 6 Reliance is the first proving ground. Hero test: rain or equipment fault, before sending a crew?
- 7 Then sell the same intelligence. Jio sends the alert; the customer decides.
- 8 The edge is integration, not the sensor.
- 9 Insurance is one application: only after the gates, a licensed insurer carries the risk.
- 10 Prices are illustrative: $\text{paid days} \times ₹300 \div 60\% \div 12$. ₹208 is not from the 3-hour rule.
- 11 90 days, three gates: coverage, accuracy, value. The audit picks the city. If yes, scale; if not, stop.
- 12 First prove the signal. Then sell the intelligence. Protection comes last.